

Experimental Characterization of a Nonlinear Electrostatic-Capacitive MEMS Resonator featuring 1:2 Internal Resonance

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Abstract— This paper describes the experimental characterization of an electrostatic-capacitive Micro Electro-Mechanical Systems (MEMS) nonlinear resonator exhibiting a 1:2 internal resonance. The proposed MEMS device is based on double-clamped parallel curved beams connected with a vertical link and has been fabricated using the ThELMA process developed by STMicroelectronics. The device geometry combined with the parallel-plate readout mechanism, enhances the nonlinear dynamic behaviour, resulting in the coupling of two in-plane vibration modes set by design in a 1:2 frequency ratio. The proposed MEMS resonator has been characterized in the frequency and time domain with and without the electrical activation of the 1:2 internal resonance. A tailored front-end circuit coupled to the resonator has been developed to measure the MEMS transfer function and the free-decay response. Experimental results have demonstrated the possibility to effectively activate the 1:2 internal resonance. The transfer function and the free-decay response highlighted the presence of two distinct harmonic contributions at 416.57 kHz and 837.81 kHz related to the two coupled vibration modes. This behaviour is consistent with the presence of nonlinear modal coupling phenomena, in which the energy introduced into the system through the excitation of one mode is partially transferred to the coupled one.

Index Terms— 1:2 internal resonance, electrostatic-capacitive, front-end circuit, MEMS, non-linear, resonator.

I. INTRODUCTION

Micro-Electro-Mechanical Systems (MEMS) are experiencing increasingly widespread adoption thanks to their small size, high performance, and low cost [1]. At the same time, new application fields are emerging, requiring devices that are more sensitive and efficient [2]. This trend is encouraging MEMS designers to investigate nonlinear phenomena, usually avoided, but now recognized as enabling both the enhancement of existing devices and the development of new operating principles. Nonlinearities improve the frequency stability over temperature fluctuations of resonant accelerometers [3] and of resonators designed for clock applications [4]. The combination of mechanical and electrostatic nonlinearities widens the linear response of MEMS sensors [5]. Among other nonlinear phenomena, internal resonance, frequently observed in nano/micro mechanical resonators [6], has been employed to stabilize the oscillation frequency of self-sustaining resonators [7], and to improve the performances of Coriolis gyroscopes [8]. Internal resonance can be also employed to generate phononic frequency combs, i.e., the mechanical counterparts of photonic frequency combs [9], which are attracting interest from the MEMS community since they can be employed as efficient sensing mechanisms. Internal resonance is defined in nonlinear vibration theory as an existing commensurate relationship between

eigenfrequencies which gives rise to mechanical coupling and energy exchange between vibrational modes [10]. In this context, this paper describes the experimental identification and characterization of the 1:2 internal resonance in a purposely designed MEMS resonator. Emphasis is placed on the experimental methodology adopted to identify the onset of internal resonance and on the analysis of the associated dynamics in both the frequency and time domains. In addition, the manuscript investigates the transient response associated with the internal resonance regime, providing experimental evidence of the coupled dynamics through time-domain measurements. This paper is organized as follows: Section II describes the proposed nonlinear MEMS resonator, Section III presents the experimental setup, Section IV the experimental results and Section V accounts for conclusions.

II. DESCRIPTION OF THE MEMS RESONATOR

The proposed MEMS device is an electrostatic-capacitive nonlinear resonator designed to guarantee the activation of the 1:2 internal resonance. Figure 1a shows the top view schematic representation of the MEMS resonator. The device embeds two parallel curved beams, set in a double-clamped configuration by means of two lateral anchors. The suspended beams, composing the MEMS shuttle, are connected by means of a rigid vertical link 27 μm long with in-plane thickness of 10 μm and are restricted to move mainly along the y-axis.

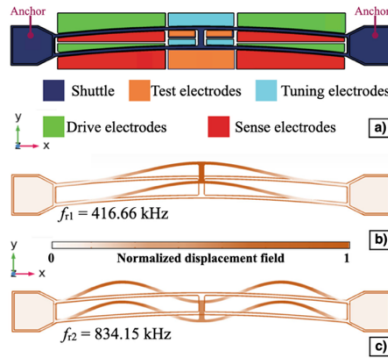


Fig. 1. Schematic top view of the proposed nonlinear electrostatic-capacitive MEMS resonator (a). Normalized displacement field of the considered in-plane vibration modes, having resonances f_{r1} (b) and f_{r2} (c) close to a 1:2 ratio.

Each beam has a cross-section of $5 \mu\text{m} \times 24 \mu\text{m}$ and a rectified length of approximately $532 \mu\text{m}$. The MEMS resonator was fabricated using the ThELMA process developed by STMicroelectronics and is made of polysilicon with a Young's modulus of 167 GPa, a Poisson's ratio of 0.22, and a density of 2330 kg/m^3 . Electrostatic actuation and capacitive sensing are achieved through parallel-plate capacitors, which fixed electrodes are shown with different colours in Fig. 1a, and are separated from the MEMS shuttle by a gap of $1.8 \mu\text{m}$. The nonlinearity of the device arises from the curved and asymmetric geometry of the structure, which guarantees non-vanishing quadratic contributions in the elastic force [10] and from the electrostatic forces generated by the actuation and sensing system employed. By applying a sinusoidal excitation voltage between the shuttle and the drive electrodes and by suitably biasing the MEMS shuttle, an electrostatic sinusoidal force is generated, allowing the mechanical vibration modes of the structure to be excited. The induced force actuates the shuttle at the drive electrodes and produces a sinusoidal displacement at sense electrodes which is converted into measurable capacitance variations. The targeted vibration modes are the first and sixth eigenmode of the MEMS device whose normalized displacement field are shown in Fig. 1b and Fig. 1c, respectively. These modes correspond to the first and second in-plane MEMS flexural mode whose resonances, neglecting fabrication tolerances, are $f_{r1} = 416.66$ kHz and $f_{r2} = 834.15$ kHz, respectively. The resonances are by design in a ratio close to 1:2 and tuning electrodes can be exploited to slightly modify

the ratio as desired through the electrostatic softening effect. By applying specific values of DC and AC voltages to the electrodes it is possible to electrically activate the 1:2 internal resonance thus favouring mechanical coupling and energy exchange between the two targeted modes. Higher-order resonance modes are located at significantly different frequencies and therefore do not contribute to the observed dynamic behaviour [10].

III. EXPERIMENTAL SETUP

To validate the working principle described in Section II the proposed MEMS resonator has been experimentally characterized both in the frequency and time domain with and without the activation of the 1:2 internal resonance. The experimental investigation was focused on excitation around f_{r1} , as the primary objective was to demonstrate the occurrence of the internal resonance phenomenon. Fig. 2a shows the electrical configuration and system implementation adopted for the frequency domain characterization. DC voltages V_{DC} and V_t have been provided to the MEMS shuttle and to the tuning electrodes to electrostatically bias the device and to induce electrostatic softening effect, respectively. The test electrodes have not been employed and thus placed at the same voltage of the shuttle V_{DC} . A sinusoidal probing signal $v_p(t)$ with rms amplitude A_p and frequency f_p has been generated by means of a gain-phase analyser (HP4194A) and provided to the drive electrodes of the MEMS resonator. The sense electrodes have been connected to the front-end circuit. Thanks to the virtual ground of the transimpedance amplifier TIA (OPA656), the sense electrode voltage is kept equal to zero and the output current $i_{out}(t)$ proportional to the capacitance variation is converted into a voltage $v_{out}(t)$. The TIA exploits a feedback resistance $R_f = 100 \text{ k}\Omega$. An inverting amplifier IA (AD817) has been embedded with $R_2 = 1 \text{ k}\Omega$ and $R_1 = 100 \Omega$ thus leading to a front-end circuit gain G :

$$G = \frac{v_{out}(t)}{i_{out}(t)} = \frac{R_f R_2}{R_1} \quad (1)$$

The output voltage $v_{out}(t)$ has been provided to the input channel of the gain-phase analyser to retrieve the transfer function $v_{out}(f_p)/v_p(f_p)$. Data are then post-processed in MATLAB to remove the feedthrough parasitic capacitance and to compensate the phase shift induced by the front-end circuit. Fig. 2b shows the electrical configuration and system implementation adopted for the time domain characterization.

A time-gated technique was employed to detect the harmonic

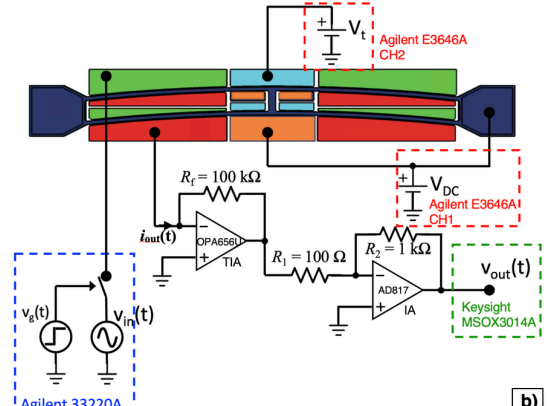
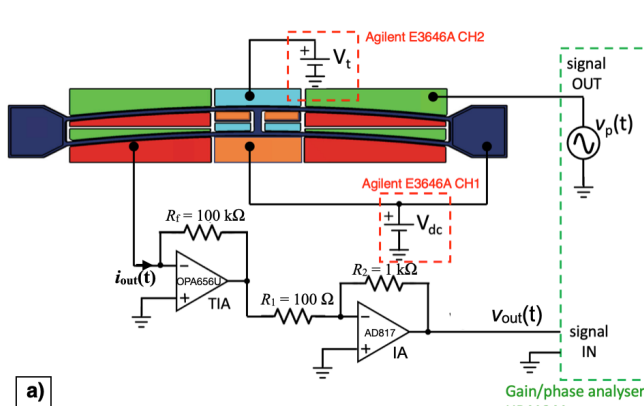


Fig. 2. Schematic representation of the adopted experimental setup to test the MEMS resonator in the frequency (a) and in the time domain (b).

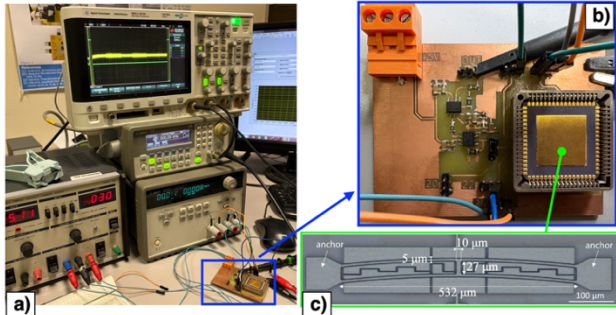


Fig. 3. Experimental setup adopted for time domain characterization (a). Enlarged view of the fabricated front-end electronic circuit (b) coupled to the MEMS device (c) enclosed in a PLCC68 package.

contributions comprising the free-decaying response of the MEMS resonator. The adopted technique consists of alternating an excitation and a detection phase. To this extent, a sinusoidal input signal $v_{in}(t)$ with rms amplitude A_{in} and frequency f_{in} is modulated by a squared signal $v_g(t)$ operating in burst mode with period $T_g = 12$ ms, amplitude $A_g = 3.3$ V and duty cycle 50%. Both signals are generated by means of a function generator (Agilent 33220A) as shown in Fig. 3a. During the excitation phase, the resonator is excited at a frequency f_{in} close to f_{r1} . During the detection phase the excitation signal is turned off and the resonator continues to oscillate freely at its resonance, progressively dissipating the accumulated energy. The motion of the resonator generates a capacitance variation between the shuttle and the sensing electrodes. The analog front-end circuit coupled to the MEMS resonator, shown in Figs. 3b,c, and the DC voltages V_{DC} and V_i were identical to that adopted for the frequency domain characterization. The MEMS device is encapsulated in near vacuum at a nominal pressure of 70 mbar in Argon and is enclosed in a PLCC68 package. The output voltage $v_{out}(t)$ has been acquired by means of the digital oscilloscope (Keysight MSOX3014) controlled by a dedicated LabVIEW interface to exploit the full data record with a sampling frequency of 100 MHz. The output signal $v_{out}(t)$ has been acquired by performing 500 consecutive measurements to enhance the signal-to-noise ratio.

IV. EXPERIMENTAL RESULTS

Figs. 4a,b show the measured $i_{out}(t)$ and $\phi(v_{out}/v_p)$ as a function of f_p swept around f_{r1} for different values of A_p . In this work only forward sweep frequency excitation have been performed. The values of A_p along with V_{DC} and V_i have been set to not activate the 1:2 internal resonance. A softening nonlinear response is obtained, as expected, with an increasing frequency shift at larger A_p .

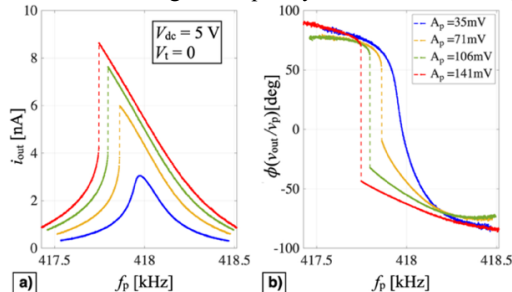


Fig. 4. Measured i_{out} (a) and $\phi(v_{out}/v_p)$ (b) as a function of frequency f_p at different A_p without the activation of the 1:2 internal resonance.

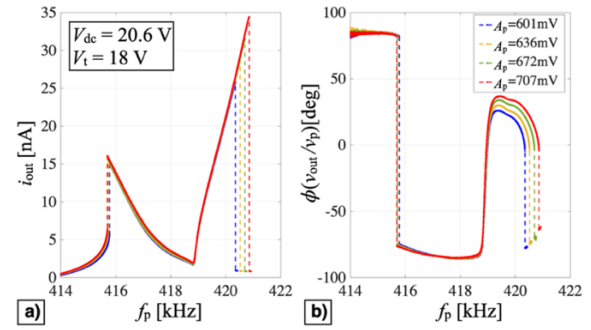


Fig. 5. Measured i_{out} (a) and $\phi(v_{out}/v_p)$ (b) as a function of frequency f_p at different A_p with the 1:2 internal resonance electrically activated.

Figs. 5a,b show the measured $i_{out}(t)$ and $\phi(v_{out}/v_p)$ with the activation of the 1:2 internal resonance. The values of V_{DC} and V_i along with A_p have been set to act on the ratio between f_{r1} and f_{r2} and to activate the nonlinear behaviour of the arch resonator, respectively. The obtained results confirm the presence of a 1:2 internal resonance, resulting in two distinct peaks associated to the coupled vibration modes, emerging at f_{r1} and $f_{r2}/2$. The measured $i_{out}(t)$ in the time domain without the 1:2 internal resonance activation has been analysed by means of a fast Fourier transform (FFT) which results are shown in Fig. 6 at different A_{in} . As expected, neglecting the DC component caused by the front-end circuit output offset, without the activation of the 1:2 internal resonance one main harmonic contribution comprising the free-decaying response appears at a frequency of 418.36 kHz and is due to the first vibration mode of the MEMS resonator. To better analyse the main harmonic contribution, $i_{out}(t)$ has been then numerically processed by applying an IIR band-pass filter around f_{r1} . Fig. 7 shows the computed $i_{out}(t)$ as a function of time t at different A_{in} . By considering the envelope of the decaying oscillations an average quality factor Q of 3310 has been derived, in agreement with theoretical expectation derived in [10]. The measured $i_{out}(t)$ in the time domain with the 1:2 internal resonance activated has been analysed by means of a FFT which results are shown in Fig. 8 at different f_{in} . The frequency f_{in} has been swept within the frequency range where the peak associated to the coupled mode appearing at $f_{r2}/2$ was observed in the achieved frequency domain results. Similarly, the values of V_{DC} and V_i along with A_{in} have been set to activate the 1:2 internal resonance. As expected, neglecting the DC component caused by the front-end circuit output offset, two main harmonics comprising the free-decaying response appear and are at the

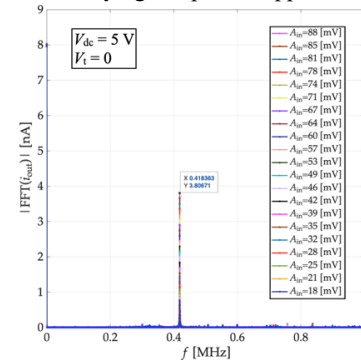


Fig. 6. Fast Fourier Transform (FFT) of the measured $i_{out}(t)$ as a function of frequency at different A_{in} without the activation of the 1:2 internal resonance.

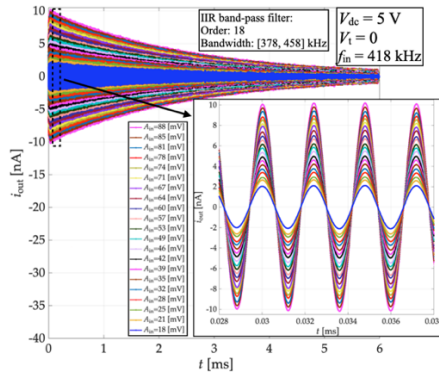


Fig. 7. Free-decaying $i_{out}(t)$ as a function of time t at different A_{in} . Enlarged time-view (inset) of $i_{out}(t)$ without the activation of the 1:2 internal resonance.

frequencies of 416.57 kHz and 837.81 kHz which are linked to the first and second in-plane MEMS flexural mode. To better analyse the two main harmonic contributions, $i_{out}(t)$ has been then numerically processed by applying an IIR high-pass filter. Fig. 9 shows the computed $i_{out}(t)$ as a function of time t at different f_{in} . The activation of the 1:2 internal resonance causes the decay response to depart from a simple exponential envelope and instead display oscillatory beats. These beats arise from coherent energy exchange between the primary mode and the parametrically excited secondary mode. The ring-down response is strongly dependent on frequency detuning, which affects the modulation depth, zero-crossing positions, and decay rate, leading to substantial amplitude variations. This sensitivity confirms that the mode coupling is governed by precise frequency matching at the 1:2 resonance condition, making the observed time-domain behaviour a clear signature of electrically activated 1:2 internal resonance. The experimental results achieved in the frequency and time domain thus confirm the nonlinear behaviour of the MEMS resonator and the possibility to activate electrically the 1:2 internal resonance.

V. CONCLUSIONS

This work presented an experimental characterization of a nonlinear electrostatic-capacitive MEMS resonator exhibiting a 1:2 internal resonance. The MEMS nonlinearities arise from the curved and asymmetric geometry of the structure, the double-clamped configuration of the beams and the electrostatic forces generated by the actuation and sensing system that exploits parallel-plate capacitors. The implemented MEMS design lead to the coupling of two in-plane

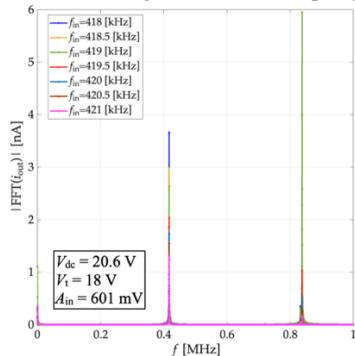


Fig. 8. Fast Fourier Transform (FFT) of the measured $i_{out}(t)$ as a function of frequency f at different f_{in} with the 1:2 internal resonance electrically activated.

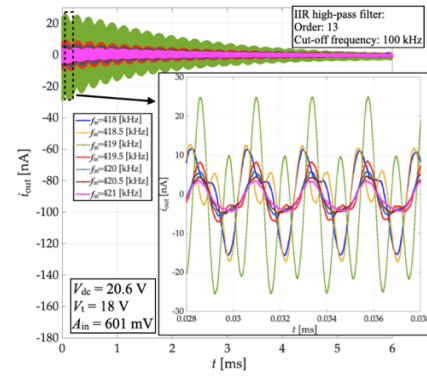


Fig. 9. Free-decaying $i_{out}(t)$ as a function of time t at different f_{in} . Enlarged time-view (inset) of $i_{out}(t)$ with the 1:2 internal resonance electrically activated.

vibration modes in a 1:2 frequency ratio. The MEMS resonator fabricated with the ThELMA process has been characterized with and without the electrical activation of the 1:2 internal resonance in both frequency and time domain. Experimental results have demonstrated the possibility to effectively activate electrically the 1:2 internal resonance. Two distinct harmonic contributions at 416.57 kHz and 837.81 kHz related to the two coupled vibration modes have been correctly measured. This behaviour aligns with nonlinear modal coupling, where energy introduced by exciting one mode is partly transferred to another coupled mode. The achieved results represent a first step toward the study of more complex phenomena, such as the generation of phononic frequency combs and toward the exploitation of model-order-reduction strategies. Such strategies are essential to support, at a low computational design cost, the next generation of nonlinear MEMS sensors.

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